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Insight into the role of intermittent microaeration in enhancing lignocellulose-containing anaerobic digestion and humic substances formation

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ABSTRACT

Anaerobic digestion of lignocellulose-containing organic matter remains challenging due to slow methane production and low conversion efficiency. To overcome these problems, intermittent microaeration has been applied to improve the performance of anaerobic digestion; however, the mechanisms are still unclear. This study investigated the formation of humic substances and their relationship with methane production under various volumes of oxygen (R1:0, R2:0.094, R3:0.188, R4:0.375 mL/(g·VS·day)). The results showed that methane production increased by 29.6 % in the R4 group, while the concentration of fulvic acid and humic acid increased by 20.4 % and 32.6 %, respectively. The fourier transform infrared spectroscopy (FTIR) analysis indicated that the formation of hydroxyl, quinone, and carboxylic acid groups in the humic substances might be influenced by microaeration. Interestingly, the concentration of humic substances was positively correlated with methane production, likely due to humic substances serving as electron shuttles to enhance the syntrophic metabolism of methanogenesis. Consistently, the electron transfer capacity, including electron accepting capacity (R1: 0.85 vs. R4: 1.86 μmol e^-/(g·VS)) and electron donating capacity (R1: 0.045 vs. R4: 0.067 μmol e^-/(g·VS)), was also enhanced with microaeration. Furthermore, the microbial community analysis revealed that lignocellulose-degrading species were enriched in the microaerobic groups. Correspondingly, the relative gene abundance of enzymes involved in the cellulose and hemicellulose metabolic pathways was higher in the group with intermittent microaeration. This study offers new perspectives on microaeration in enhancing anaerobic digestion and supports its potential for widespread application in sustainable waste-to-energy systems.

1. Introduction

Anaerobic digestion is widely regarded as a sustainable and eco-friendly method to convert biodegradable organic matter into renewable energy (methane) [1,2]. Straw is one of the main types of agricultural wastes with an annual production of approximately 8.6 billion tons in China [3]. However, anaerobic digestion of straw encounters obstacles owing to the primary composition of lignocellulose, which consists of 40–50 % cellulose, 20–30 % hemicellulose, and 15–25 % lignin [4]. When the lignocellulosic structure of organic matter serves as the primary carbon source for anaerobic digestion, it is difficult for anaerobes to decompose these complex matters, resulting in the slow organic conversion and the low methane production [5].

Recently, it was reported that a small quantity of oxygen or air (DO 0.1–1 mg/L) benefited anaerobic digestion and increased methane production [6]. Air or oxygen injected into an anaerobic digestion is believed to promote the growth of facultative bacteria that secrete extracellular hydrolytic enzymes such as cellulases to decompose cellulose [7]. The facultative bacteria consume oxygen in a certain extent to relieve its harm to obligate anaerobic methanogens [6]. Furthermore, it has been reported that *Methanosarcina* and *Methanobacterium* can produce antioxidative enzymes to survive under high DO level [8]. In high-solids anaerobic digestion of cow manure, air mixing system (37.5 (mL/L·min)) increased methane production by 13 % [9]. The air-mixed full-scale reactor sustained high biogas yields over a year of continuous operation [10]. The enhanced anaerobic digestion with microaeration was mainly ascribed to the enrichment of hydrolytic and fermentative bacteria to increase microbial diversity. However, the underlying

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mechanisms remain unclear.

Humic substances, including fulvic acid (FA) and humic acid (HA), are generated during anaerobic digestion via microbial decomposition and polymerization [11]. During this process, the organic matter is gradually broken down and transformed into more stable high-molecular-weight compounds. For example, when lignocellulose-containing substrates are used as a carbon source, they can provide structural precursors (lignin theory) or low-molecular-weight organics (polyphenol theory) that contribute to the formation of humic substances. Humic substances have varying effects on different stages of anaerobic digestion [12]. Specifically, it can enhance solubilization, acidogenesis, and acetogenesis, while accelerating or inhibiting hydrolysis and methanogenesis [13]. These effects may depend on the concentration of humic substances. For example, at higher concentrations (2–4 g/L), HA might inhibit methane production by negatively affecting methanogenesis [14]. In contrast, at lower concentrations (< 1 g/L), HA can act as an electron shuttle because of the presence of quinone groups, thereby increased methane production by 16–81 % [14,15]. In fact, oxygen plays an important role in humic substances formation by microbial activities during aerobic composting [16]. Similarly, the addition of trace amounts of oxygen might influence the formation of humic substances, thereby affecting methane production. For example, Zhu et al. reported that microaeration during the anaerobic digestion of corn straw increased soluble HAs concentration [17]. However, further studies are still needed to deepen the understanding how microaeration affects humic substances formation and methanogenesis.

In this study, corn straw and waste activated sludge (WAS) were co-digested with intermittent microaeration at various air volumes. The main objectives of this study were to: 1) evaluate the impact of microaeration on anaerobic digestion performance, including methane production and organic matter removal efficiency; 2) investigate the effects of microaeration on humic substances formation – measured by HA and FA concentrations, and two dimensional correlation analysis fourier transform infrared spectroscopy (2D-cos-FTIR) – as well as on humification, assessed by HA/FA ratio and gemination index (GI); 3) elucidate the effects of humic substances on microbial electron transfer systems via differential pulse voltammetry (DPV), in situ FTIR, cyclic voltammetry (CV), and electron donating capacity (EDC)/electron accepting capacity (EAC); and 4) explore the possible mechanisms of microaeration effects on anaerobic digestion from the aspects of the microbial community and metagenomic analysis.

2. Materials and methods

2.1. Substrate and inoculum

Corn straw was purchased from a farm (Lianyungang, China), chopped, ground into powder, and screened to a particle size of approximately 0.15 mm. The waste activated sludge (WAS) was collected from the secondary sedimentation tank of a wastewater treatment plant in Dalian, China. The inoculum sludge was taken from two anaerobic reactors: one from an anaerobic digester treating cow manure in Hangzhou, China (Inoculum HZ) and another from an anaerobic digester that used WAS and kitchen waste as substrates in Dalian, China (Inoculum DL). The primary characteristics of the substrates and inoculum are listed in the Table 1.

Table 1
Characteristics of substrates and inoculum.

	CS	WAS	Inoculum HZ	Inoculum DL
TS (%)	93.8 ± 0.1	6.4 ± 0.0	6.1 ± 0.3	4.0 ± 0.2
VS (% of TS)	72.2 ± 0.1	61.5 ± 1.8	54.5 ± 2.8	82.6 ± 2.3
C (% of TS)	33.7 ± 0.0	33.4 ± 0.5	–	–
N (% of TS)	0.9 ± 0.0	5.6 ± 0.1	–	–
H (% of TS)	3.9 ± 0.7	4.9 ± 0.1	–	–

2.2. Batch experiment

The batch experiments were conducted in anaerobic reactors with an effective volume of 200 mL, and each group had three parallel reactors. At the beginning of the experiment, 10 g of corn straw powder, 40 g of WAS, 30 mL of Inoculum HZ, and 20 mL of Inoculum DL were mixed in each reactor, followed by the addition of Milli-Q water to achieve a total volume of 200 mL. The main characteristics of the reactors were as follows: total solids (TS) 81.0 g/L, volatile solids (VS) 56.0 g/L, total chemical oxygen demand (TCOD) 80.37 g/L, and pH 7.33. All the reactors were purged with nitrogen gas for 20 min to create anaerobic conditions. During the experiment, different volumes of air - 0, 5, 10, and 20 mL (0, 0.094, 0.188, 0.375 mL/(g·VS·day) O_2) were injected into each reactor daily and set as groups **R1, R2, R3,** and **R4** (Fig. S1). All reactors were placed in a constant temperature room at 37 °C and shaken at 120 rpm.

2.3. Physicochemical analysis

The TS, VS, and COD were measured according to standard methods [18]. The C, N, and O in the corn straw were analyzed using an elemental analyzer (Elementar, vario EL cube, China). pH was measured using a pH meter (STARTER, 3100, USA). The gas composition (methane and carbon dioxide) was evaluated using gas chromatography (Tianmei, GC7900, China) with a thermal conductivity detector (TCD). At the end of the experiment, sludge samples from the four groups were freeze-dried and mixed with KBr for FTIR analysis using a spectrometer (ThermoFisher, 6700, USA). The 2D-cos-FTIR data was performed using the 2D Shige software. Data processing was conducted using Origin 2018, while *t*-test for data significance was conducted using the SPSS 31. The GI was evaluated based on a previous study [19].

Humic substances were extracted from sludge samples using a modified method [20,21] as follows: 1 g of freeze-dried samples was mixed with Milli-Q water at a 1:10 (m/v) ratio, and the pH was adjusted to 1.0 using 6 M HCl. The mixture was shaken at 40 °C for 1 h, centrifuged at 10000 rpm for 25 min, and the supernatant was filtered through Whatman GF/C glass-fiber filters to obtain extraction FA1. The precipitate was then adjusted to pH 12.0 with 0.1 M $NaOH/Na_4P_2O_7$ and shaken under N_2 conditions for 4 h. The mixture was centrifuged at 10,000 rpm for 25 min, and the supernatant was filtered through Whatman GF/C glass fiber filters. The filtrate was acidified to pH 1.0 with 6 M HCl to form a suspension. This suspension was centrifuged at 10,000 rpm for 15 min. The supernatant obtained was designated as extracted FA2, while the precipitate was collected as extracted HA. The extracted HA was dissolved in 0.1 M KCl, and additional solid KCl was added to adjust the K^+ concentration to 0.3 M. The solution was then centrifuged at 8000 rpm, and 6 M HCl was added to the supernatant to induce reprecipitation. The precipitate was collected by centrifugation, the supernatant was discarded, and the precipitate was resuspended in a mixture of 0.1 M HCl and 0.3 M HF. The suspension was shaken overnight, transferred to a dialysis bag, dialyzed against Milli-Q water, and then freeze-dried. In addition, FA1 and FA2 were purified via XAD-8 resin adsorption and then freeze-dried. The total FA concentration was calculated as the sum of the FA1 and FA2.

2.4. Electrochemical analysis

Both CV and DPV were measured using a three-electrode microbial electrochemical system connected to an electrochemical workstation (Chenhua, CHI650B, China). The working electrode was a glassy carbon electrode (1 cm × 1 cm), the counter electrode was a platinum sheet electrode (1 cm × 1 cm), and the reference electrode was an Ag/AgCl electrode. 1 × PBS solution was used as the electrolyte, and the headspace of the electrochemical system was purged with nitrogen. For CV analysis, the potential range was set from-0.8 to 0.8 V, with scan rates of 50 mV/s. The EAC and EDC were analyzed as described in a previous

study [22]. The parameters for DPV measurement were: potential ranges from −0.8 to 0.4 V or 0.4 V to −0.8 V, amplitude 0.05 V, pulse width 0.2 s, and pulse period 0.5 s.

Electrochemical in situ FTIR was detected as reported by Yu et al. [23]. A thin-layer configuration equipped with a CaF_2 window was positioned vertically outside the infrared spectroscopy chamber. The working, counter, and reference electrodes were platinum, platinum sheet, and Ag/AgCl electrodes, respectively. 1 × PBS solution was used as the electrolyte. The various potentials of the working electrode were controlled by a workstation (Corrtest, CS1008, China) in the range of −0.9, −0.8, −0.6, −0.4, −0.2, 0, 0.2, and 0.4 V, respectively. The FTIR spectra were obtained from 400 to 4000 cm^{-1} at the different potentials.

2.5. Microbial community and metagenomic analysis

At the end of the experiment, 2 mL of samples from each group were taken for DNA extraction using the E.Z.N.A™ Mag-Bind Soil DNA Kit. After DNA concentration measurements and quality control, PCR amplification was performed. For bacteria, the V3-V4 region was amplified using the primers 340F and 805R. For archaea, nested PCR targeting the V3-V4 region was processed with two rounds of amplification. The primers used in the first round were 340F and 805R, while those used in the second round were 349F and 806R. The pooled and purified PCR products were sequenced using an Illumina Hiseq2000 sequencer.

For metagenomic analysis, total DNA was extracted from each sample and its quantity and purity were measured before constructing a library. Quality-controlled libraries were sequenced using an Illumina NovaSeq 6000 high-throughput sequencing platform. Fastp 0.23.2 software was used for quality control of the sequenced data, followed by de novo assembly of clean data using the MEGAHIT software (v1.2.9) [24]. N connections were then segmented to obtain N-free scaffolds, which were processed using Prodigal 2.6.3 [25] to predict open reading frames (ORFs). The predicted ORFs from each sample were combined and clustered using the Linclust software to generate a non-redundant gene catalog [26]. Finally, the abundance of lignocellulose-degrading related gene abundance was analyzed using the KEGG and CAZyme databases. The raw 16S rRNA and metagenomic sequencing data were deposited in the NCBI BioProject database under SRA accession number PRJNA1287080.

3. Results and discussion

3.1. Reactor performance

The 40-day batch experiment showed that methane production increased with the daily addition of different volumes of oxygen (Fig. 1A). There was no significant difference in the accumulative methane production among the four reactors during the initial 10 days. Afterwards, the accumulative methane production significantly increased in the R3 (0.188 mL/(g·VS·day) O_2) and R4 (0.375 mL/(g·VS·day) O_2) groups during the last 30 days ($P < 0.05$). It might be attributed to the longer time required for the hydrolytic acidogenesis processes in this co-digestion system, as straw was the main carbon source. At the end of the experiment, the accumulative methane production in the R2, R3, and R4 groups were 60.2 ± 0.6 mL/(g·VS), 67.5 ± 2.8 mL/(g·VS), and 77.0 ± 5.4 mL/(g·VS), which were 1.3 %, 13.6 %, and 29.6 % higher than those of control group, respectively. The addition of more oxygen has a positive effect on methane production, suggesting that intermittent microaeration could facilitate the hydrolytic acidogenesis of lignocellulose, thereby providing more suitable substrates for methanogenesis. Similarly, Zhu et al. also reported that anaerobic digestion of straw for methane production was increased when the oxygen concentration lower than 0.8 mL/(g·VS) [17].

As shown in Fig. 1B, accumulative carbon dioxide production exhibited a similar trend to that of methane production, which was also

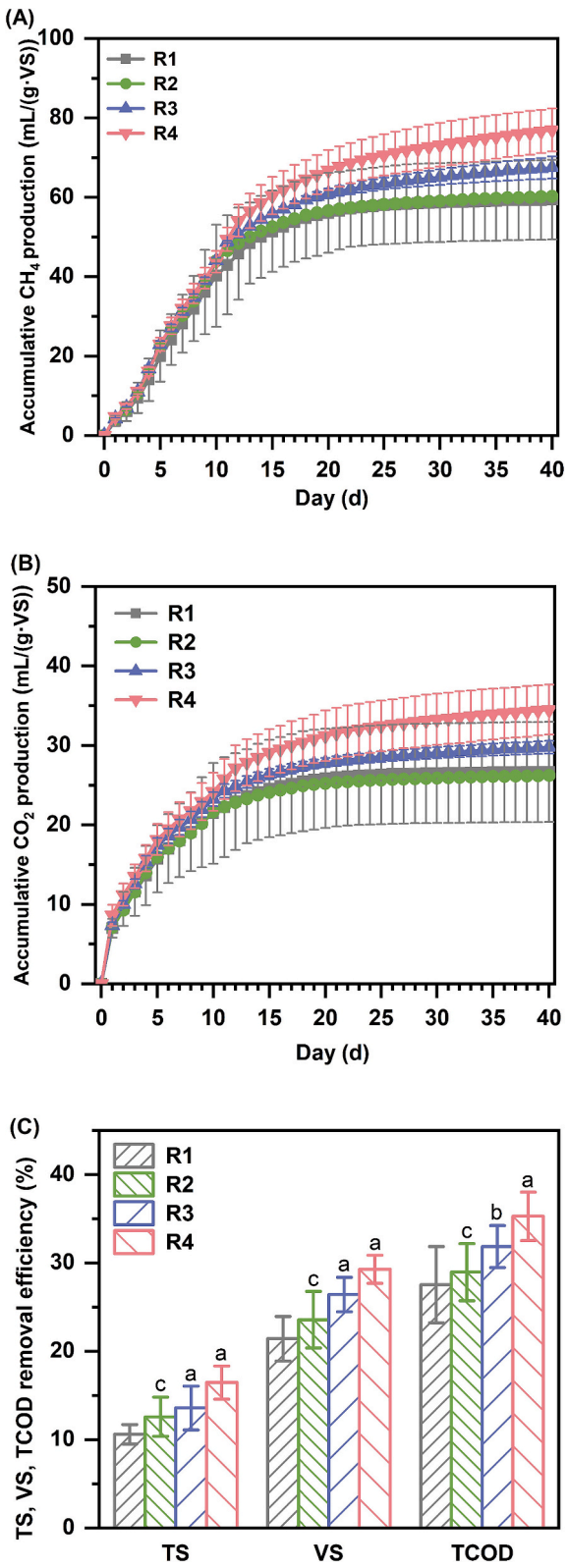


Fig. 1. Anaerobic reactor performance with different volumes of air: Accumulative CH_4 production (A), Accumulative CO_2 production (B), TS, VS, and TCOD removal efficiency at the end of the experiment (C). 'a' denotes a two-tailed independent *t*-test with statistical significance ($P < 0.05$); 'b' denotes a one-tailed independent t-test with statistical significance ($P < 0.05$); and 'c' denotes a non-significant result from an independent t-test ($P > 0.05$).

enhanced by the daily addition of oxygen. On day 40, the accumulative carbon dioxide production in the R1 group was 26.7 ± 6.3 mL/(g·VS), while it was 11.4 % and 29.3 % higher in the R3 (29.8 ± 0.8 mL/(g·VS)) and R4 (34.5 ± 3.2 mL/(g·VS)) groups compared to the R1 group, respectively. The increased carbon dioxide production also demonstrates that the decomposition of lignocellulose-containing organic matter was accelerated by intermittent microaeration. Consistently, the TS, VS, and TCOD removal efficiencies were also the highest in the R4 group, reaching 16.5 %, 29.3 %, and 35.3 %, respectively (Fig. 1C).

3.2. Humic substances formation

Humic substances are produced through the transformation of lignocellulosic structures by microorganisms [27]. As shown in Fig. 2A, peaks at 3400 cm^{-1} (O—H stretching), 2930 cm^{-1} (asymmetric C—H stretching of CH_2 group), 2850 cm^{-1} (symmetric C—H stretching of CH_2 group), 1650 cm^{-1} (C═O stretching in quinone), 1450 cm^{-1} (aliphatic C—H deformation) and 1050 cm^{-1} (C—O stretching in the polysaccharides or polysaccharide-like substances) became more obvious in the groups with intermittent microaeration [28]. Among them, the peak at 3400 cm^{-1} was related to hydroxyl groups, and the peak at 1650 cm^{-1} was associated with the quinone group, both of which were likely originated from humic substances, suggesting a possible correlation with microaeration and the formation of humic substances.

The 1800–800 cm^{-1} region of the FTIR spectra was analyzed in Fig. 2B and C, as these regions were related to carboxylic acids, carbohydrates, and aliphatic groups [29]. There were five auto-peaks (903, 1297, 1487, 1624, and 1771 cm^{-1}), indicating the spectral intensities detected in the synchronous map. The peak at 903 cm^{-1} was attributed to the aromatic fingerprint region, whereas the peak at 1297 cm^{-1} was associated with C—C stretching in the quinone rings [30,31]. The peaks at 1487 cm^{-1} and 1624 cm^{-1} correspond to C═C stretching in the aromatic rings and C═O stretching in the quinone rings, respectively [29]. Specifically, the most significant changes were observed at 1771 cm^{-1}, which were assigned to the carboxylic acid group [32], suggesting that intermittent microaeration might influence the formation of carboxylic acids in humic substances. Moreover, cross-peaks can reflect the correlation between the intensity changes at two different spectral coordinates. Together with asynchronous maps, the reaction order of humic substances functional groups was 1297 > 1487 > 1771 > 1624 cm^{-1}.

3.3. Quantity and quality of humic substances

FA and HA are the two main components of humic substances [11]. At the end of the experiment, the humic substances concentrations were measured. The results indicate that the groups with intermittent microaeration significantly enhanced the formation of humic substances ($P < 0.05$) (Fig. 3A). FA is a water-soluble compound with a relatively simple structure, mainly consisting of short-chain and cyclic compounds and serving as a precursor for HA formation. The concentration of FA in the four groups was as follows: R1 (34.6 ± 1.2 mg/kg) < R2 (37.9 ± 1.5 mg/kg) < R3 (40.2 ± 5.2 mg/kg) < R4 (41.7 ± 1.7 mg/kg). HA has a relatively complex structure and contains a high proportion of aromatic substances. The concentration of HA in the R1 group without microaeration was 4.6 mg/kg, whereas in the R2, R3, and R4 groups, it was 19.0 %, 23.1 %, and 32.6 % higher than that in the R1 group, respectively.

On the other hand, the HA/FA ratio is a key indicator of the degree of humification, with a higher ratio suggesting a more mature and stable transformation of organic matter [33]. The HA/FA ratios in the four groups were R1 (0.134) < R3 (0.141) < R2 (0.145) < R4 (0.147) (Fig. 3B). In addition, GI is also an indicator of the maturity and quality of humic substances, reflecting their phytotoxicity and bioavailability [34]. The GI of the R1 group was 62.1 % and that of the R2 group was 67.7 %, both lower than the national standard of China (>70 %), whereas it was 77.2 % in the R3 group and 73.8 % in the R4 group (Fig. 3C). Intermittent microaeration improved both humic substances formation and methane production. A low concentration of humic substances can enhance methane production by promoting direct

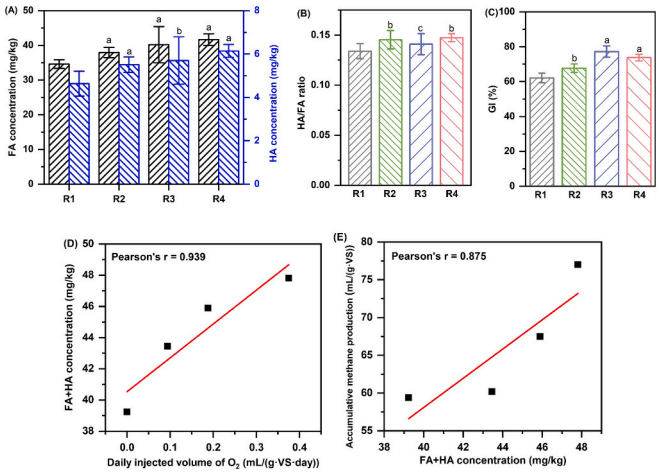


Fig. 3. FA and HA concentration (A), HA/FA ratio (B), GI (C), correlation between humic substances (FA + HA) concentration and daily injected oxygen volume (D), and correlation between methane production and humic substances concentration (E). 'a' denotes a two-tailed independent *t*-test with statistical significance ($P < 0.05$); 'b' denotes a one-tailed independent t-test with statistical significance ($P < 0.05$); and 'c' denotes a non-significant result from an independent t-test ($P > 0.05$).

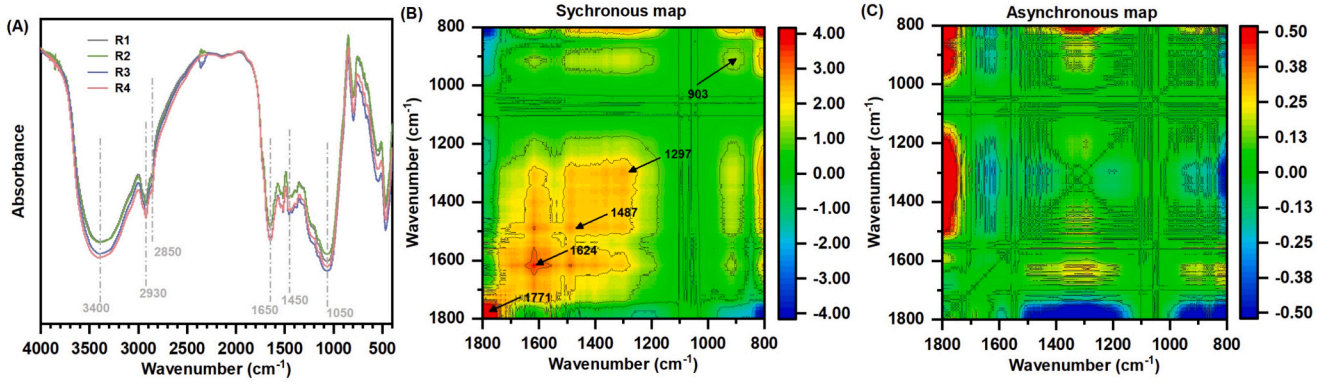


Fig. 2. FTIR spectra (A), synchronous (B), and asynchronous (C) 2D correlation maps generated from the 1800–800 cm^{-1} region of the FTIR spectra of sludge samples from four groups at the end of the experiment.

interspecies electron transfer between electron-donating microbes and methanogens or by acting as electron shuttles [15]. The concentration of humic substances generated in this study was within the low concentration range, and thus had positive effects on methane production.

To further investigate the correlation between the volume of intermittent oxygen addition and methane production, a linear fitting analysis was conducted in Fig. 3D and E. The results showed a strong positive correlation between the volume of intermittent microaeration and the concentration of humic substances ($y = 21.67x - 40.54$, $R^2 = 0.882$), with a Pearson coefficient of 0.939. Additionally, the Pearson correlation coefficient for the linear fit between humic substances concentration and accumulative methane production ($y = 1.93x - 19.26$, $R^2 = 0.765$) was 0.875, indicating a positive relationship.

3.4. Effects of humic substances on electron transfer capacity

DPV analysis was used to analyze the redox peaks formed during electron transfer in the systems. Three oxidation peaks were detected in all four groups, with potentials of −0.533 V, −0.335 V, and − 0.046 V (vs Ag/AgCl, R4 group) (Fig. 4A). Similarly, three reduction peaks were also observed, with potentials of −0.636 V, −0.357 V and −0.164 V (vs Ag/AgCl, R4 group) (Fig. 4B). Owing to the complexity of the microbial reaction, it was difficult to accurately identify the specific substances corresponding to each redox peak. Nevertheless, humic substances also contained multiple redox peaks, including peaks at −0.16 V and −0.58 V, which were similar to peaks observed in this study [35]. Noteworthily, as the volume of added oxygen increased, the corresponding redox peaks became more obvious and displayed a negative shift compared to the control group, indicating an improvement in electron transfer [36].

The redox process of the functional moieties (quinones) of humic substances can be characterized by electrochemical in situ FTIR [31]. As shown in Fig. 4C, when the electrode potential shifted from −0.9 V to 0.4 V, several potentials depended upward bands could be detected. The upward bands at 1476, 1626 and 1720 cm^{-1} were assigned to vibration of C═C, C═O stretch in the quinone rings and C═O in carboxyl groups [31]. These bands might be related to electron transfer and oxidation of humic substances in the R1 and R4 groups, whereas the R4 group had more obvious band intensities than the R1 group.

The capacitance, which is the integral area of the CV curve, reflects the electron transfer ability between microorganisms [37]. As shown in Fig. 4D, at a scan rate of 50 mV/s, the capacitances of the four groups were as follows: R2 (2.36 mF) < R1 (2.38 mF) < R3 (2.42 mF) < R4

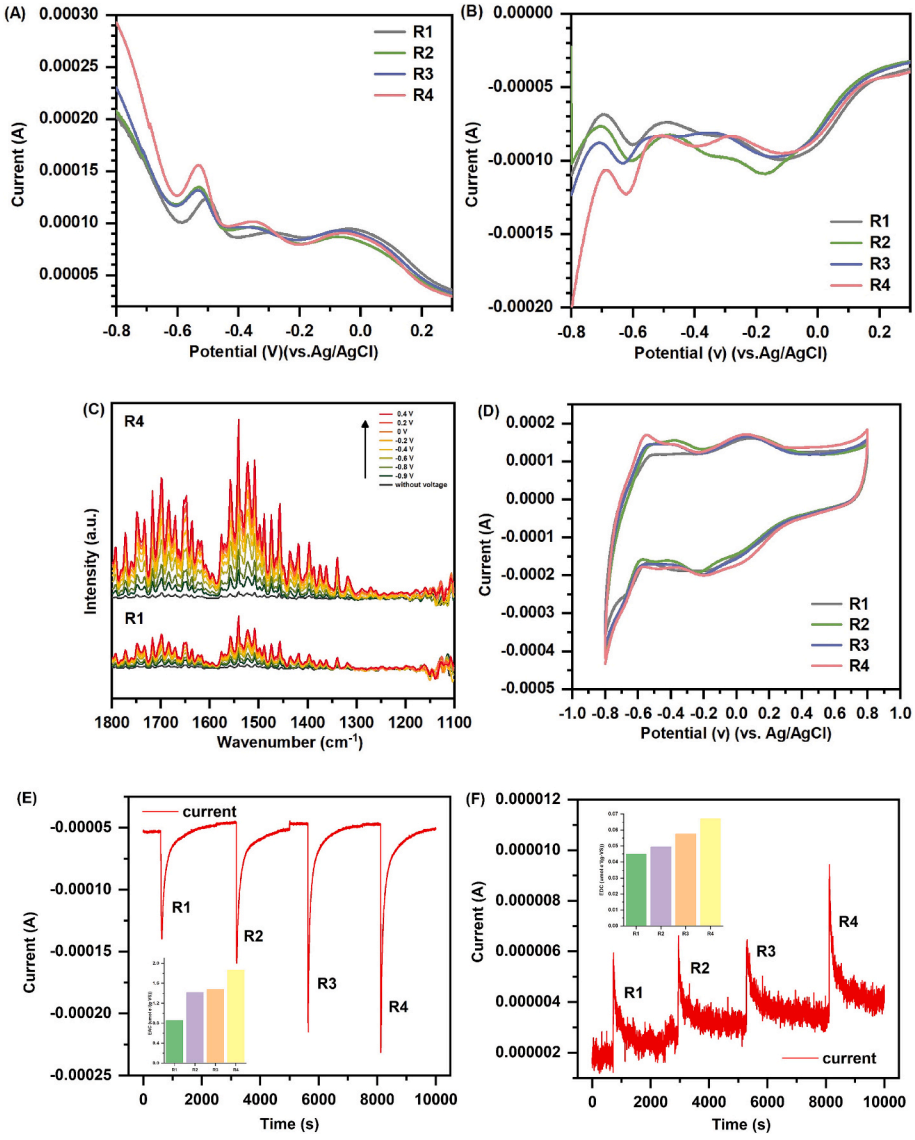


Fig. 4. Electrochemical analysis: DPV analysis (A and B); in situ FTIR of sludge samples from R1 and R4 groups at 1200–2000 cm^{-1} (C); CV analysis at scan rate of 50 mV/s (D); Electron accepting capacity (E), and Electron donating capacity (F).

(2.63 mF). A higher capacitance indicates a greater ability to charge or discharge electrons during electron transfer. Besides, EAC and EDC were increased in the groups with microaeration. As shown in Fig. 4E, the EAC of the R1 group was 0.85 μmol e^-/(g·VS), while it increased to 1.41 μmol e^-/(g·VS) in R2 group, 1.48 μmol e^-/(g·VS) in R3 group, and 1.86 μmol e^-/(g·VS) in R4 group. Similarly, the EDC of the four groups were as follows: R1 (0.045 μmol e^-/(g·VS)) < R2 (0.049 μmol e^-/(g·VS)) < R3 (0.058 μmol e^-/(g·VS)) < R4 (0.067 μmol e^-/(g·VS)) (Fig. 4F). These results indicate that intermittent microaeration with different oxygen volumes might improve the microbial electron transfer capacity by promoting the production of more humic substances.

3.5. Microbial community analysis

As shown in Fig. 5A, the bacterial community at the genus level revealed that the lignocellulose-degrading microorganisms were enriched in the groups with microaeration. Specifically, *norank_Prolixibacteraceae*, *norank_Anaerolineaceae*, *Fastidiosipila*, *norank_Bacteroidales and norank_Christensenellaceae* were identified as lignocellulose-degrading species. The total relative abundance of these species was 45.8 % in the R1 group, 47.8 % in the R2 group, 49.6 % in the R3 group, and 54.8 % in the R4 group. Among them, *norank_Prolixibacteraceae* (family Prolixibacteraceae), which is associated with lignin degradation [38], was the most dominant lignocellulose-degrading genus, with a relative abundance of 17.3 % (R1), 25.5 % (R2), 28.4 % (R3), and 21.8 % (R4). Members of the Prolixibacteraceae family are capable of metabolizing a wide range of carbohydrates [39]. Another dominant lignocellulose-degrading genus in the four groups was *norank_Anaerolineaceae* (family Anaerolineaceae). *Norank_Anaerolineaceae* has been reported to secrete extracellular cellulase to degrade cellulose and enhance hydrolysis, but lacks genes related to carbohydrate-binding modules (CBMs) [40]. The highest relative abundance of *Norank_Anaerolineaceae* was observed in the R4 group (22.8 %). Furthermore, *Fastidiosipila*, another lignocellulose-degrading microorganism, accounted for 3.6 %, 5.1 %, 5.0 %, and 4.8 % in the R1-R4 groups, respectively. *Fastidiosipila* plays an important role in the conversion of complex macromolecular organic matter into VFAs and carbon dioxide [41]. In addition, exoelectrogens capable of extracellular electron transfer (e.g., *Syntrophomonas*, *Thermomonas*) were enriched in the groups with microaeration [42,43]. The relative abundance of *Syntrophomonas* was 0.68 %, 1.03 %, 1.25 %, and 0.72 % in the R1-R4 groups, respectively.

The dominant species in the archaeal community of four groups at the genus level were *Methanosaeta*, *Methanospirillum*, *Methanomassiliicoccus*, *Candidatus_Methanofastidiosum*, and *norank_Bathyarchaeia* (Fig. 5B). Specifically, *Methanosaeta* was the most dominant species, accounting for 56–57 % in the all of four groups. Generally, *Methanosaeta* is an acetoclastic species and can obtain electrons from electron-donating species via DIET to facilitate propionate degradation [44]. In addition, *norank_Bathyarchaeia*, which is associated with lignin degradation, showed lowest relative abundance in the R4 group [45]. This might be attributed to the most of the lignocellulose had already been degraded by lignocellulose-degrading bacteria in the R4 group.

3.6. Metagenomic analysis

Carbohydrate-active enzymes (CAZymes) play a crucial role in lignocellulose degradation. It can be seen from Fig. 6A, glycoside hydrolases (GHs), glycosyltransferases (GTs), and CBMs accounted for the highest proportion of CAZymes in R1 and R4 groups. GHs are a class of enzymes that catalyze the hydrolysis of glycosidic bonds (covalent bonds linking sugar molecules to other molecules) and break down complex carbohydrates into smaller sugars or monosaccharides. GHs are a class of enzymes that catalyze the hydrolysis of glycosidic bonds (covalent bonds between sugar molecules and other molecules), breaking down complex carbohydrates into smaller sugars or monosaccharides [46]. GTs catalyze the formation of glycosidic bonds between the activated sugar groups and acceptor molecules [47]. CBMs are non-catalytic domains that bind specifically to carbohydrate substrates (e.g., cellulose and hemicellulose), positioning the enzyme catalytic domain near the substrate to enhance efficiency [48]. The relative abundance of these three types of enzymes was slightly higher in the R4 group than in the R1 group.

Fig. 6B and C further analyze the relative gene abundance of enzymes involved in the cellulose and hemicellulose metabolic pathways, showing higher levels in the group with intermittent microaeration. For example, GH43, including enzymes like β-D-xylosidase, can debranch and hydrolyze hemicellulose [48]. The relative abundance was 0.0028 in the R4 group, which was 7.7 % higher than in the R1 group. In addition, the relative gene abundance of CBM3 (0.00029 vs 0.00025), CBM6 (0.00112 vs 0.00098), and CBM32 (0.0011 vs 0.0010) were also increased in the R4 group. CBM3 is associated with the cellulose enzyme catalytic domain of GH, CBM6 is involved in the degradation of cellulose, xylan, and glucan, and CBM32 is distantly related to CBM6 [49,50].

The pathway for anaerobic digestion of complex organic matter

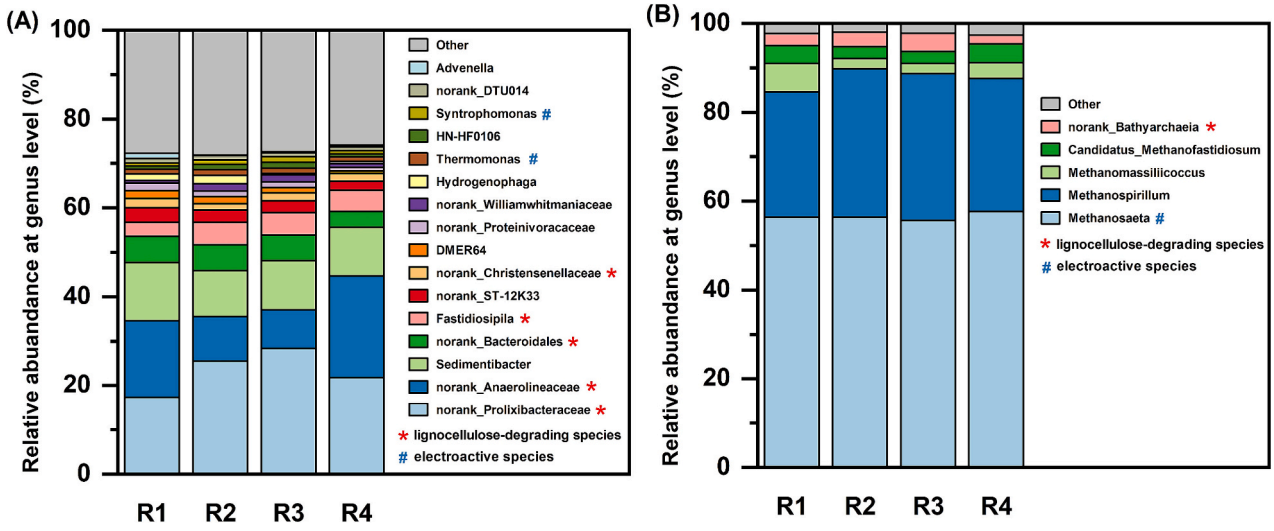


Fig. 5. Microbial community analysis: relative abundance of bacteria at genus (relative abundance >1 % at least in one sample) level (A) and archaea at genus level (B).

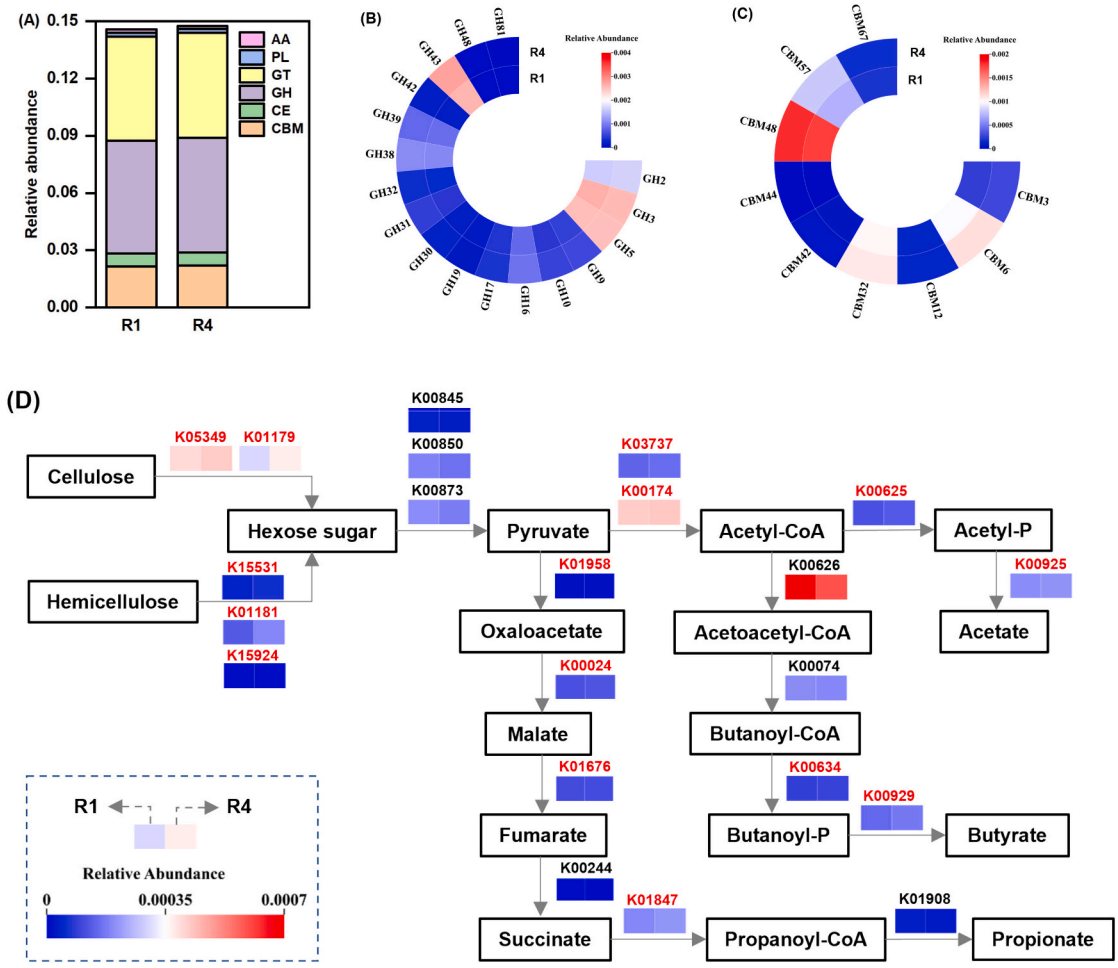


Fig. 6. Metagenomic analysis of samples from R1 and R4 groups: Relative abundance of CAZyme at class level (A); CAZyme at family level related to cellulose and hemicellulose degradation in GHs (B) and CBM (C) ; pathways and relative abundance of genes assigned to hydrolysis and acidogenesis of corn straw based on KEGG database (D).

containing lignocellulose was as follows [51]. First, cellulose/hemicellulose was degraded into oligosaccharides, converted into pyruvate, and finally transformed into VFAs such as acetate, propionate, and butyrate. The relative abundance of the key genes associated with each metabolic pathway are shown in Fig. 6D. The relative abundance of genes related to cellulose degradation (bglX (K05349) and cel (K01179)), as well as genes related to hemicellulose degradation (rexA (K15531), xynA (K01181), and xynC (K15924)), increased in the R4 group, which was consistent with the CAZyme results (Fig. 6A–C). Although the process of converting oligosaccharides into pyruvate did not improve, the subsequent conversion of pyruvate into VFAs, especially acetate (pyruvate → acetyl-CoA → acetyl-P → acetate) and propionate (pyruvate → oxaloacetate → malate → fumarate, succinate → propanoyl-CoA) was enhanced.

According to the polyphenol–protein theory, some small molecular compounds can condense to form humic substances through microbial metabolism and enzymatic activity during anaerobic digestion [20]. This process can be divided into the intracellular synthesis of polyphenolic compounds and extracellular condensation of these polyphenolic compounds to form HA and FA [20,52]. The M00004 (PPP pathway), M00022 (shikimate pathway), M00024 (phenylalanine biosynthesis), and M00025 (tyrosine biosynthesis) modules were associated with the formation of intracellular polyphenols and the generation of humic substances [53,54]. Both M00024 and M00025 belonged to the ko00400 (phenylalanine, tyrosine, and tryptophan biosynthesis) metabolic pathway, and the relative abundance of related genes in the R4 group were slightly higher than those in the R1 group (Table S1). After the intracellular polyphenolic substances were released into the extracellular to condense FA and HA by enzymes such as polyphenol oxidase and ubiquinone oxidoreductase [20]. EC1.3.5.1 and EC7.2.1.1 belonged to the ubiquinone oxidoreductase. The relative gene abundance was 0.00031and 0.00060 in the R4 group, which increased by 3.0 % and 15.6 %, respectively, compared to the R1 group. This is consistent with the FA and HA concentrations (Fig. 3).

3.7. Implication

In this study, intermittent microaeration was applied to promote the anaerobic co-digestion of corn straw and waste activated sludge, resulting in enhanced methane production and increased formation of humic substances. This strategy provides a sustainable solution for the efficient utilization of lignocellulose-containing organic waste. Microaeration supports the activity of facultative microorganisms involved in lignocellulose degradation [7] and influences the structural transformation of lignocellulose, thereby contributing to humic substance formation—either via structural precursors (lignin theory) or low-molecular-weight organics (polyphenol theory). Consequently, the generated humic substances not only enhance organic matter stabilization but also facilitate microbial electron transfer, which contributes to more efficient methanogenesis. These findings offer practical insights for the treatment of a broad range of lignocellulose-containing substrates, including crop residues, cattle manure, and agro-industrial

byproducts. Importantly, balancing oxygen dosage and microbial activity is critical for optimizing system efficiency. Although microaeration improved both substrate degradation and electron transfer capacity, further investigation is needed to determine the precise oxygen threshold that maximizes energy recovery while avoiding oxidative inhibition.

4. Conclusion

The present study demonstrated that intermittent microaeration enhanced both methane production and humic substances formation during the anaerobic digestion of lignocellulose-containing substrates. FTIR analysis indicated that microaeration might influence the formation of functional groups such as hydroxyl, quinone, and carboxylic acid groups, which are key structural components of the humic substances. In addition, the quality of humic substances (such as the HA/FA ratio and GI) was also improved under microaerobic conditions, especially at an oxygen dosage of 0.375 mL/(g·VS·day). The formation of humic substances was positively correlated with accumulative methane production. Moreover, the electron transfer capacity was also enhanced in the microaerobic groups, since humic substances can act as electron shuttles. Furthermore, microbial community and metagenomic analysis further revealed that microaeration enriched lignocellulose-degrading microorganisms and upregulated genes associated with cellulose and hemicellulose degradation.

CRediT authorship contribution statement

Zhen Jin: Writing – original draft, Data curation. **Qilin Yu:** Methodology. **Zhiqiang Zhao:** Conceptualization. **Yaobin Zhang:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2025.166500.

Data availability

Data will be made available on request.

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